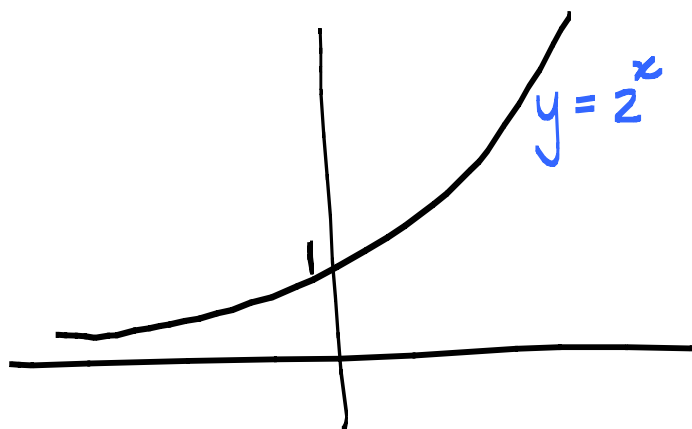


Unit 1.7 Exponential functions p45

Note Title

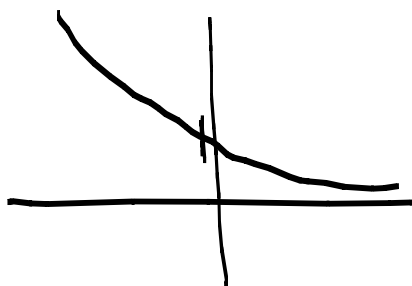
2014/02/08



In general:

$$0 < a < 1$$

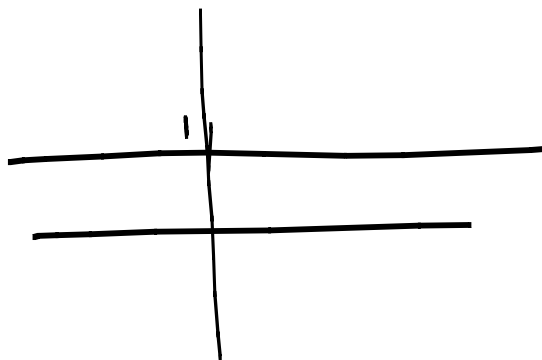
$$y = \left(\frac{1}{2}\right)^x = \left(2^{-1}\right)^x = 2^{-x}$$



$$y = a^x$$

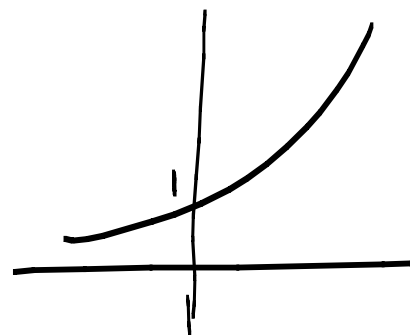
$$a = 1$$

$$y = 1^x$$



$$a > 1$$

$$y = 2^x$$



Laws of Exponents If a and b are positive numbers and x and y are any real numbers, then

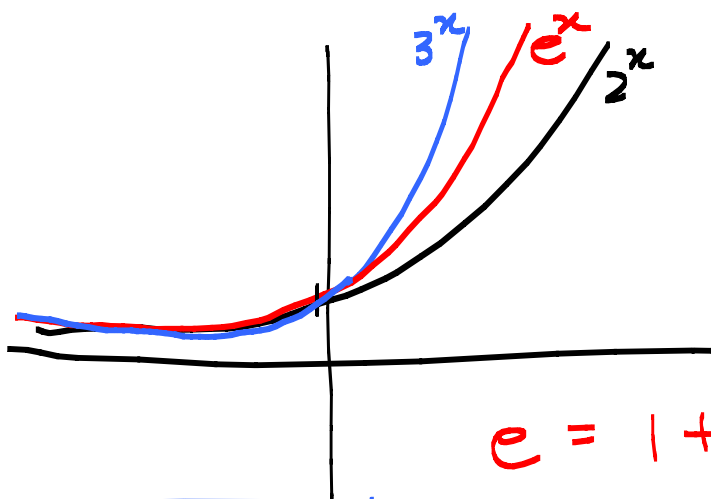
1. $a^{x+y} = a^x a^y$

2. $a^{x-y} = \frac{a^x}{a^y}$

3. $(a^x)^y = a^{xy}$

4. $(ab)^x = a^x b^x$

Sketch $f(x) = 2^x$ and $g(x) = 3^x$ on the same set of axes.



$$2 < e < 3$$

$$e = 2,71828 \dots$$

Euler's number.

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots$$

$$2! = 2 \cdot 1 = 2$$

$$4! = 4 \cdot 3 \cdot 2 \cdot 1$$

$$1 + e^{i\pi} = 0$$

Maths
2.5

The number e !

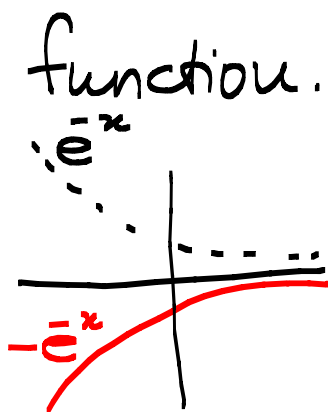
Euler's number.

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots$$

$$= 2,71828$$

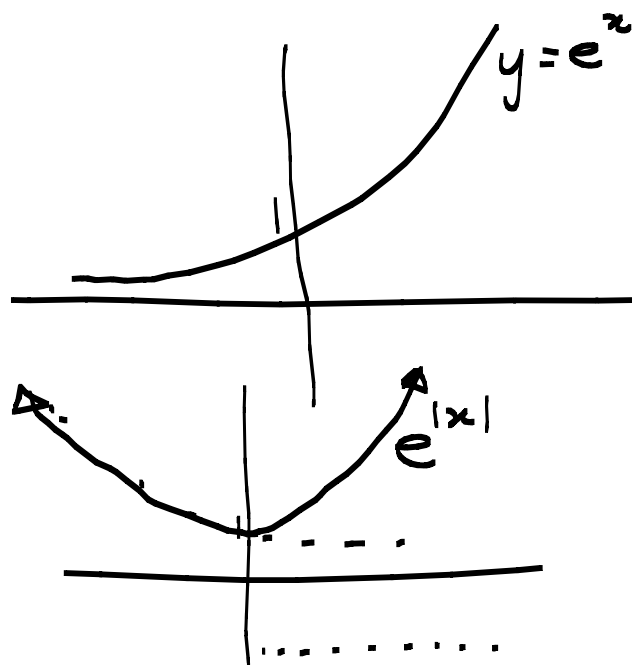
Natural exponential function.

$$y = -e^{-x}$$



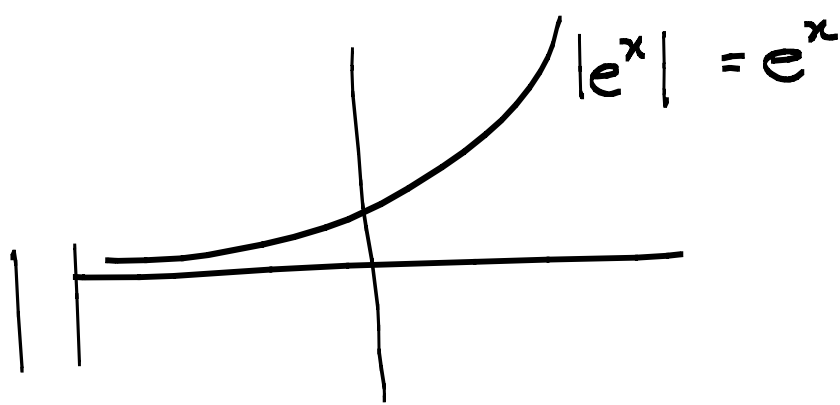
$$y = \begin{cases} e^x & \text{if } x \geq 0 \\ e^{-x} & \text{if } x < 0 \end{cases}$$

$$y = e^x$$



$$y = |e^x|$$

$$\boxed{e^x > 0}$$



Rules: Simplify $\frac{(e^x \cdot e^y)^4}{(e^e)^5}$

$$= \frac{(e^{x+y})^4}{e^{5e}} = e^{4x+4y-5e}$$

① Find the domain of $f(x) = \frac{1}{1+e^x}$

$$\sqrt{\quad}$$

consider

$$1+e^x = 0$$

$$\Rightarrow e^x = -1$$

not possible.

$$D_f = \mathbb{R}$$

② $g(t) = \sin(e^{-t})$

$$D_g = \mathbb{R}$$

$$(f \circ h)(t)$$

$$= f(h(t)) \quad D_h = \mathbb{R}$$

$$h(t) = e^{-t} \quad D_f = \mathbb{R}$$

$$f(t) = \sin(e^{-t})$$

$$(3) \quad g(t) = \sqrt{1 - e^t}$$

$$1 - e^t \geq 0 \quad \Rightarrow \quad e^t \leq 1 \quad ??$$

Inverse of e !!

To be continued.....